Uber Fare Estimator (Regression)

Technical Report

Machine Learning, Chapter 14 Capstone Project (Project 1)

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1. Abstract

This project predicts the fare of a New York taxi trip from where and when it starts and ends. Fare is a continuous number, so this is a regression problem. I used the Uber fares dataset from Kaggle (200,000 trips from 2009 to 2015). After cleaning, 191,801 trips were used. I built features such as the straight-line (Haversine) distance, time of day and distance to the three big airports. A Linear Regression baseline reached a test RMSE of 3.832 USD and R^2 of 0.831. A Random Forest tuned with RandomizedSearchCV reached a test RMSE of 2.998 USD, MAE of 1.562 USD and R^2 of 0.897, about 22% lower RMSE than the baseline. On average the estimate is off by about 1.56 USD.

2. Data Exploration

The file uber.csv has 200,000 rows and 9 columns. I used 7 of them: fare_amount, pickup_datetime, pickup and dropoff longitude and latitude, and passenger_count. Only one row has missing dropoff coordinates. The raw data is dirty: the minimum fare is -52 USD, the maximum is 499 USD, some coordinates are impossible (for example a longitude of -3356 or a latitude of 1644) and passenger_count ranges from 0 to 208.

Cleaning rules:

- Remove rows with missing values and duplicates.
- Keep fares above 0 and below 200 USD (the 200 USD limit is my own judgement).
- Keep passenger counts from 1 to 6.
- Keep only trips inside a box around New York City (latitude 40.5 to 41.0, longitude -74.3 to -73.6).
- Remove trips shorter than 0.1 km.

After the first cleaning steps 194,915 of 200,000 rows remained (97.5%); after removing near-zero distances 191,801 rows were left.

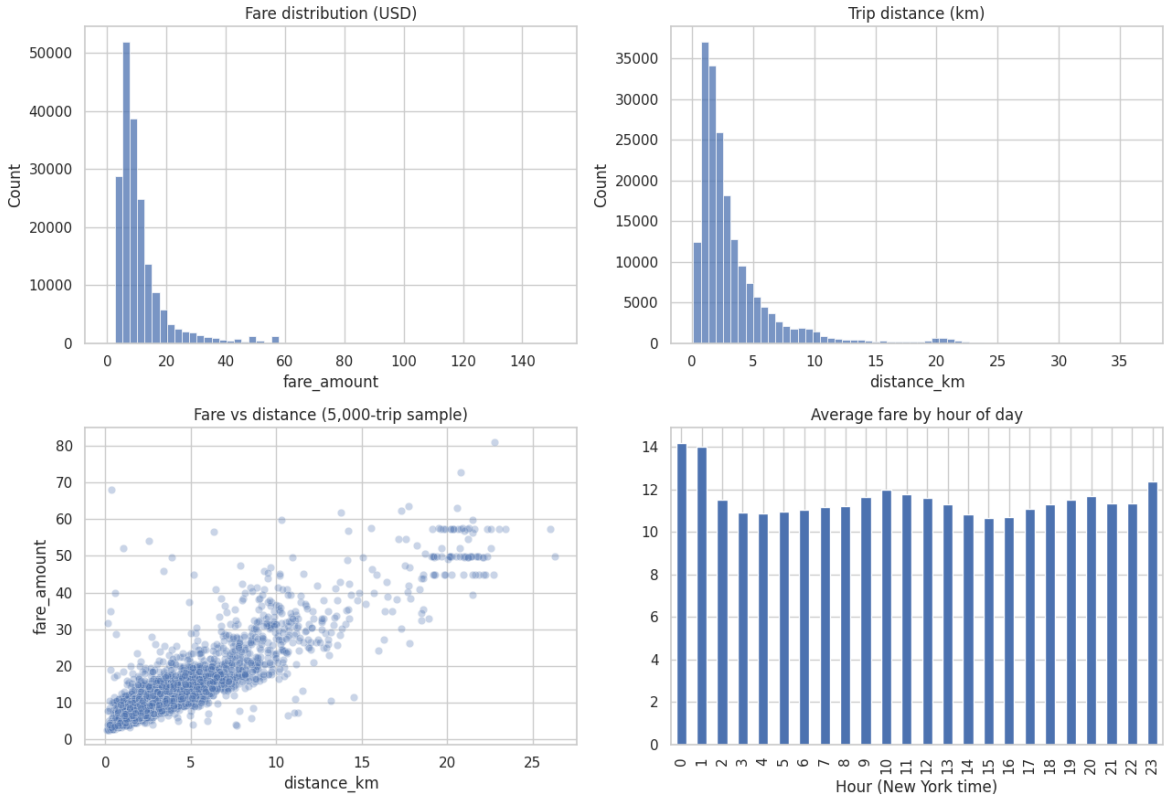


Figure 1. Fare and distance distributions, fare against distance, and average fare by hour.

Main observations:

- Distance has the strongest correlation with fare (0.899). Pickup distance to JFK (-0.459) and pickup longitude (0.440) come next.
- Fares and distances are right-skewed: most trips are short and cheap, with a long tail of expensive trips.
- Some trips of about 20 km have nearly flat fares of 45 to 57 USD. These may be airport flat fares (my guess, not checked).
- Average fare is highest around midnight (about 14 USD at hours 0 and 1) and about 11 to 12 USD at other hours. Hour has only a small effect.

3. Methodology

Features (17). Pickup and dropoff coordinates, passenger count, Haversine distance in km, hour, day of week, month, year and a weekend flag (in New York local time), and the distance of pickup and dropoff to JFK, LaGuardia and Newark airports.

Split. The data was split before fitting anything: 80% train (153,440 trips) and 20% test (38,361 trips), random_state 42. A random split is reasonable because single trips are independent of each other.

Preventing leakage. For the baseline, the StandardScaler is inside a Pipeline, so it learns its mean and standard deviation from the training data only. Tuning used cross-validation on training data only.

Models. The baseline is Linear Regression. The complex model is a Random Forest, which averages many decision trees and can learn non-linear patterns that a straight line cannot. I trained a default forest (100 trees) and a tuned forest.

Tuning. RandomizedSearchCV with 8 random trials and 3-fold cross-validation, scored by RMSE, run on a random sample of 20,000 training rows to save time. The search covered n_estimators, max_depth, min_samples_leaf and max_features. Best settings: n_estimators 150, max_depth None, min_samples_leaf 2, max_features sqrt (cross-validated RMSE 3.172). The tuned forest was then retrained on the full training set.

Metrics. RMSE (typical error in USD, punishes big mistakes), MAE (average absolute error in USD) and R^2 (share of fare variation explained).

4. Results

Model	Train RMSE	Test RMSE	Test MAE	Test R^2
Linear Regression (baseline)	3.919	3.832	2.123	0.831
Random Forest (default)	1.216	3.085	1.627	0.891
Random Forest (tuned)	**1.765**	**2.998**	**1.562**	**0.897**

Table 1. Results on the training set and the unseen test set (errors in USD).

The tuned Random Forest is the best model. Its test RMSE is 2.998 against 3.832 for the baseline, about 22% lower, and R^2 rises from 0.831 to 0.897. Tuning gave only a small gain over the default forest (3.085). The train RMSE (1.765) is lower than the test RMSE (2.998), so the forest overfits somewhat, but the tuned model overfits less than the default one (1.216 vs 3.085).

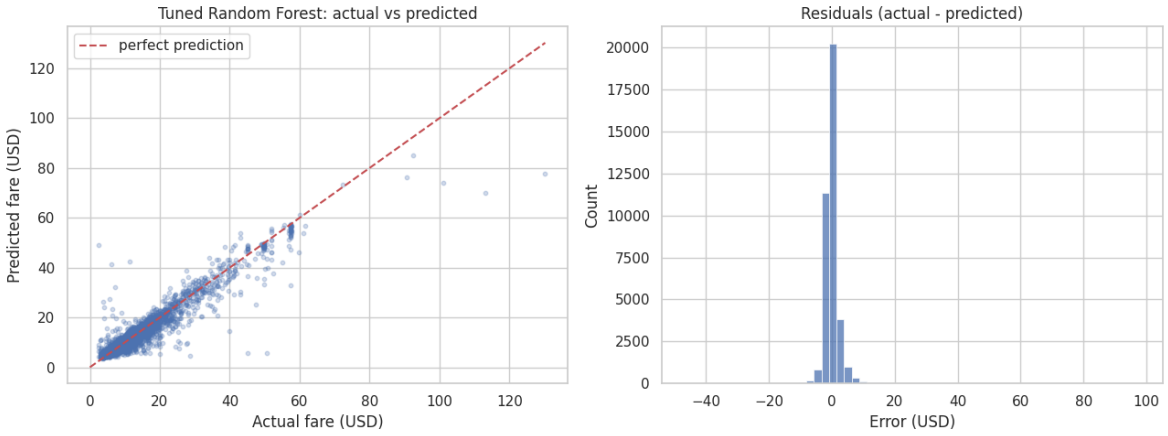


Figure 2. Tuned Random Forest: actual vs predicted fare (left) and residuals (right).

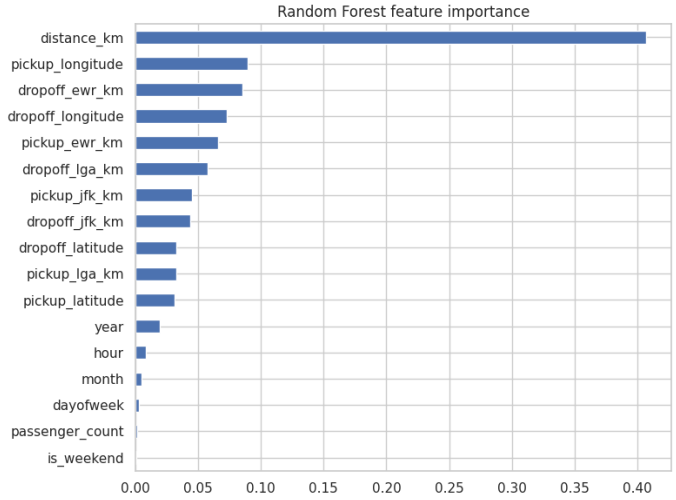


Figure 3. Random Forest feature importance. Distance is the most important feature (about 0.41).

Demo. For a trip from Times Square to JFK (3 June 2015, 16:30 UTC, one passenger) the model estimates a fare of $55.63. This is the model's estimate; I did not compare it with a real fare.

5. Conclusion

Best model. The tuned Random Forest (test RMSE 2.998, MAE 1.562, R^2 0.897) beat the Linear Regression baseline (RMSE 3.832, R^2 0.831) by about 22% in RMSE.

Meaning in practice. On average the estimated fare is off by about 1.56 USD. Bigger mistakes also happen, as the RMSE of about 3 USD shows.

Limitations. There is some overfitting (train RMSE 1.765 vs test 2.998). The data has no traffic or weather information, covers several years of changing fares, and the tuning used only 8 random trials on a 20,000-row sample. Long trips such as airport rides are harder to predict. The Haversine distance is a straight line, not the real road distance.

Future work. Add traffic and weather data, try XGBoost, search more settings, and deploy the model with FastAPI.